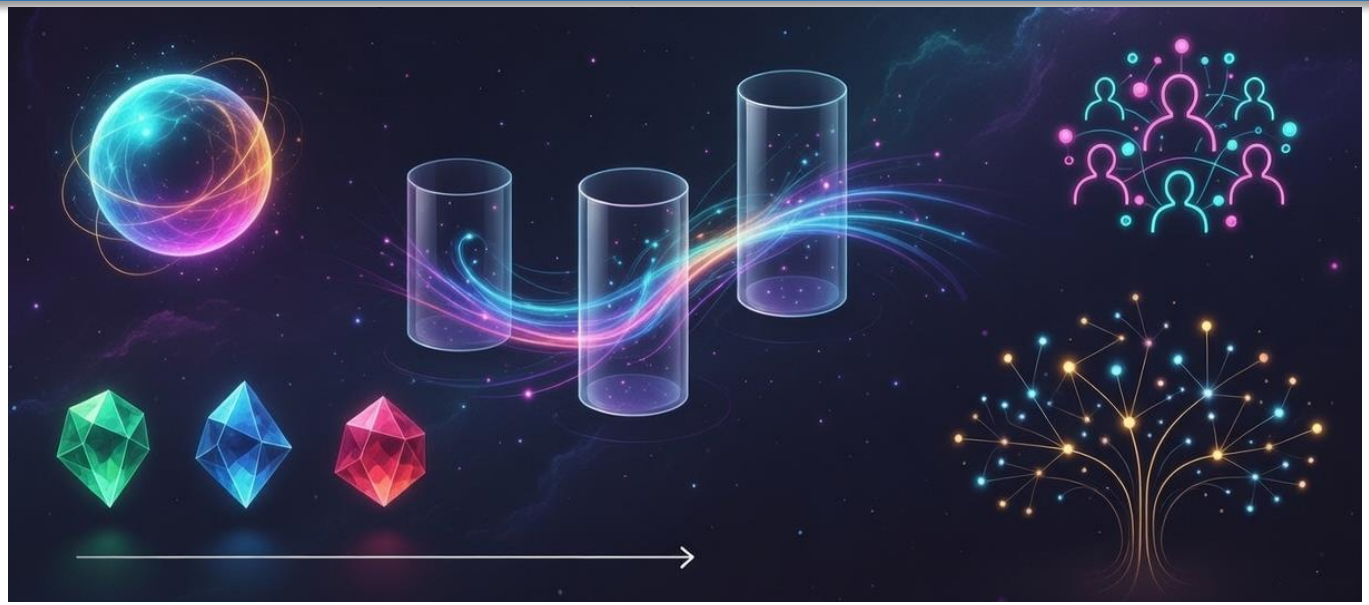
CCS1510 - Introduction to Groupwork Project



Assignment

- Do “reverse engineering” (design by examples) by investigating a specific existing well-known site you choose (from a list of predefined sites)
- Design the database, providing the design assumptions / business logic
- Implement the database
- Populate database with sample data
- Answer questions about data by writing useful queries in SQL

Choose a Web Site

1. AirBnb.com

2. Imdb.com

3. Formula1.com

Project Scope

- These sites are huge
  - It is out-of-scope to model the whole web site for this project
- You are provided with some assumptions
  - Check with the lecturer to ensure you don't do too much or too little

Project Details & Milestones

- Groups of 4 people
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MILESTONE	DESCRIPTION	DELIVERABLES	TIMELINE
0	Form Groups	Submit Declaration Form	*Week 3*
1	Design	Submit 1. Informal Report 2. AI Journal	*Week 6*
2	Implementation	Submit: 1. Informal Report 2. SQL Code 3. AI Journal	*Week 10*
3	Presentation	1. Submit Presentation Slides 2. Attend Presentation (MANDATORY)	*Week 12*
4	Peer Review	Peer Assessment Form	*Week 12*

AI Use

- Required to use generative AI tools in specific ways and to document usage transparently
- You may use general-purpose AI tools (e.g Copilot, ChatGPT, Claude, Gemini) and/or Data Design tools under some rules
  - MUST DO
    - Log your AI use in an AI Journal
    - Critically Evaluate AI output; you are responsible for correctness
  - NEVER DO
    - Copy Paste AI-generated content without understanding it
    - Use AI to fabricate results (e.g fake images)

What do I use AI For?

- AI can be used for:
  - Brainstorming domain entities, attributes, business rules
  - Creating / critiquing ER diagrams and schemas
  - Generating realistic sample data
  - Drafting or improving SQL queries, explanations, and report prose
  - Creating Presentation Slides

Undeclared or prohibited AI use will be treated as an academic integrity issue according to university regulations

AI Journal

- Maintain AI Journal (e.g. Google Sheets)
  - Maintained across M1 and M2
- For every significant AI interaction you use for the project, you must record the following (e.g. as separate columns in Google Sheets):
  1. Tool name (e.g. ChatGPT, Copilot, Visual Paradigm AI)
  2. Purpose (e.g. “check our ERD for redundancy”)
  3. Prompt used (what you asked)
  4. Response - you can select only Key parts of the response (copy or screenshot is fine)
  5. Your evaluation
     - What was useful?
     - What was wrong or misleading?
     - What did you keep or change, and why?

Deliverables – M1 Informal Report

- Word or PDF Format
  - No hard limit (minimum or maximum) on the number of words
- Should contain the following:
  - **Introduction -** Description of the site and what needs / scenarios it covers
  - **Assumptions -** Extra assumptions (not ones already mentioned) to keep scope limited
  - **Design -** Overview of main entities and relationships, including ERD
  - **AI Use -** What AI proposed, what/why did you change, what have you learned about ER modeling and AI's strengths/weaknesses
- Important Note
  - You should NOT be explaining every single shape and line in the ERD
  - Explain important aspects only, e.g. why you picked a specific relationship to be one-to-one instead of one-to-many, etc

Deliverables – M2 Informal Report

- Word or PDF Format
  - No hard limit (minimum or maximum) on the number of words
- Should be based on the one you wrote for Milestone 1
  - Modified after the feedback you receive
  - Note: A problematic ERD will produce an erroneous Relational Schema and in turn this will produce an undesirable implementation of the database
- Additionally, include the following:
  - Relational Schema, as mapped from the ERD
  - Sample Data Specification / Insertion Strategy
    - The way / reasoning of how you populated the database with sample data to make it "adequate" for testing your queries (e.g how many rows in each table and why)

Deliverables – M2 Database Implementation

- Implement on Microsoft SQL Server

- Submit SQL code files (.sql)
  1. Schema (CREATE TABLE etc)
  2. Sample data generation (INSERTS)
  3. SQL Code for Queries (described in the handout, different for each of the sites)

- How to save some time regarding Sample Data
  - Use “minimum” amount of data to test your queries (mention in the report how many rows per table and why)
  - During your data entry, you don’t have to filling in nullable fields that will not take part in any of your queries

What you need to do now

- Form groups and select a web site
  - Submit form by end of week
- Attend classes and do all of the ERD examples